1.9.3 Bras and Kets

As we have seen, the complex numbers have a dual version: in the form of complex conjugate numbers. In the same way, a complex vector space has a dual version that is essentially the complex conjugate vector space. For every ket-vector |*A*>, there is a “bra” vector in the dual space, denoted by <*A*|. Why the strange terms *bra* and *ket*? Shortly, we will define inner products of bras and kets, using expressions like <*B*|*A*> to form *bra-kets* or brackets. Inner products are extremely important in the mathematical machinery of quantum mechanics, and for characterizing vector spaces in general.

Bra vectors satisfy the same axioms as the ket-vectors, but there are two things to keep in mind about the correspondence between kets and bras:

1. Suppose <*A*| is the bra corresponding to the ket |*A*>, and <*B*| is the bra corresponding to the ket |*B*>. Then the bra corresponding to

$$|A\rangle + |B\rangle$$

is

$$\langle A| + \langle B|.$$

2. If *z* is a complex number, then it is *not* true that the bra corresponding to *z*|*A*> is <*A*|*z*. You have to remember to complex-conjugate. Thus, the bra corresponding to

$$z|A\rangle$$

is

$$\langle A|z^*.$$